

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Computer Vision – 21CSE454T

UNIT 2: SHAPES AND REGIONS | SESSION 8: BOUNDARY DESCRIPTORS

SLO 1: Explain different types of boundary descriptors used for shape representation in binary images.

ACTIVITY 1 – FIND THE STARTING PIXEL SOLUTION

Matrix representation (using 0-indexed coordinates (row, col)):

```
Row 0: 0 0 0 0 0
Row 1: 0 [1] 1 1 0    (<-- Top-most, left-most boundary pixel at index
(1,1))
Row 2: 0 1 0 1 0
Row 3: 0 1 1 1 0
Row 4: 0 0 0 0 0
```

Starting coordinate: **(1, 1)** (or row 1, column 1)

ACTIVITY 2 – CHAIN CODE DIRECTION MAP REFERENCE

Direction	Code
Right	0
Down-Right	1
Down	2
Down-Left	3
Left	4
Up-Left	5
Up	6
Up-Right	7

ACTIVITY 3 – TRACE THE BOUNDARY SOLUTIONS

Starting clockwise traversal from **(1, 1)** around the hollow square contour:

- Step 1 (Move Right from (1,1) to (1,2)): **0**
- Step 2 (Move Right from (1,2) to (1,3)): **0**
- Step 3 (Move Down from (1,3) to (2,3)): **2**
- Step 4 (Move Down from (2,3) to (3,3)): **2**
- Step 5 (Move Left from (3,3) to (3,2)): **4**
- Step 6 (Move Left from (3,2) to (3,1)): **4**

First six chain codes:

Step 1 → **0**

Step 2 → **0**

Step 3 → **2**

Step 4 → **2**

Step 5 → **4**

Step 6 → **4**